MULTI-MODAL ONCOLOGY AI ANALYSIS REPORT

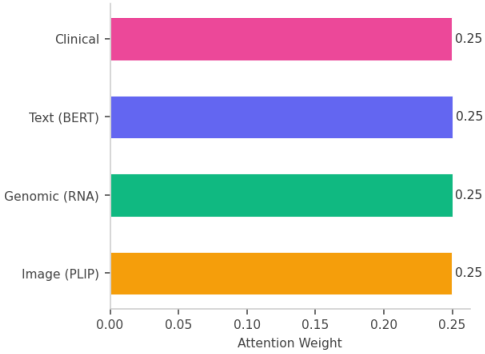
Breast Cancer Precision Medicine Pipeline | 2026-04-14 14:26 | Patient: TCGA-AO-A0JF

SUBTYPE: HR+ | Confidence: 100% | Survival Risk: LOW (4th pct)

AI CLASSIFICATION: HR+ (100% confidence) | Survival Risk: LOW (4th percentile)

Prediction

Modality Contributions

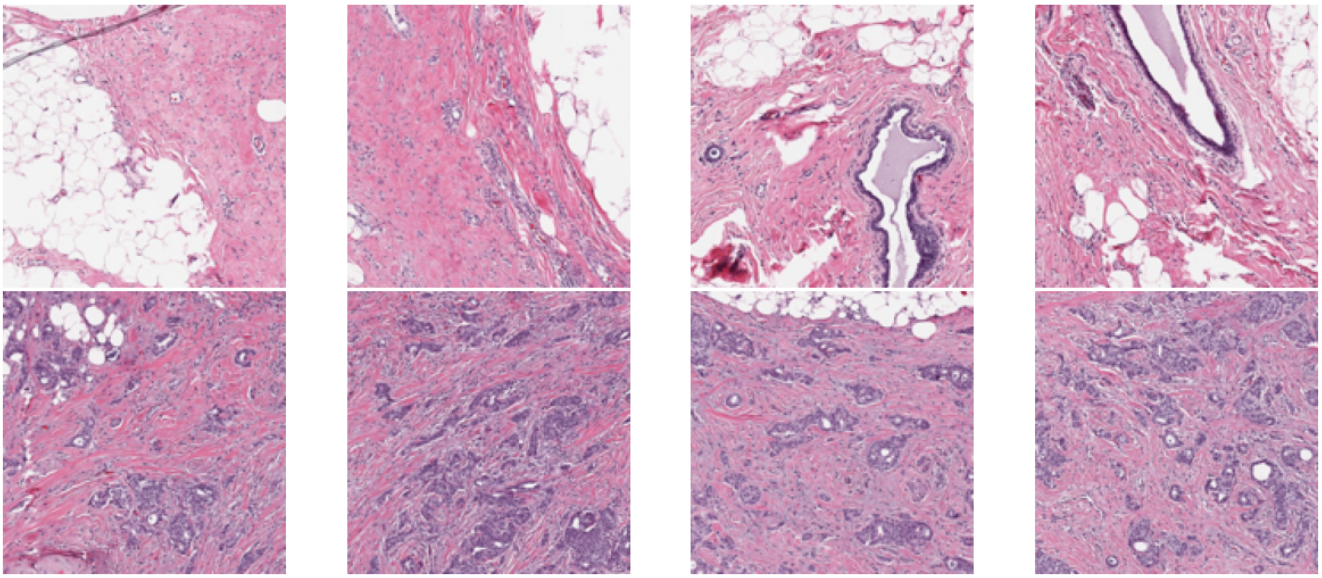


Patient Summary

	Value
Patient ID	TCGA-AO-A0JF
True Subtype	HR+
Predicted	HR+ (100%)
Survival Risk	LOW (4th pct)
ER / PR / HER2	Pos / Pos / Neg
Stage	Stage IIA
Age	68
Top Drivers	RNA_ESR1, RNA_PGR, RNA_CCND1,

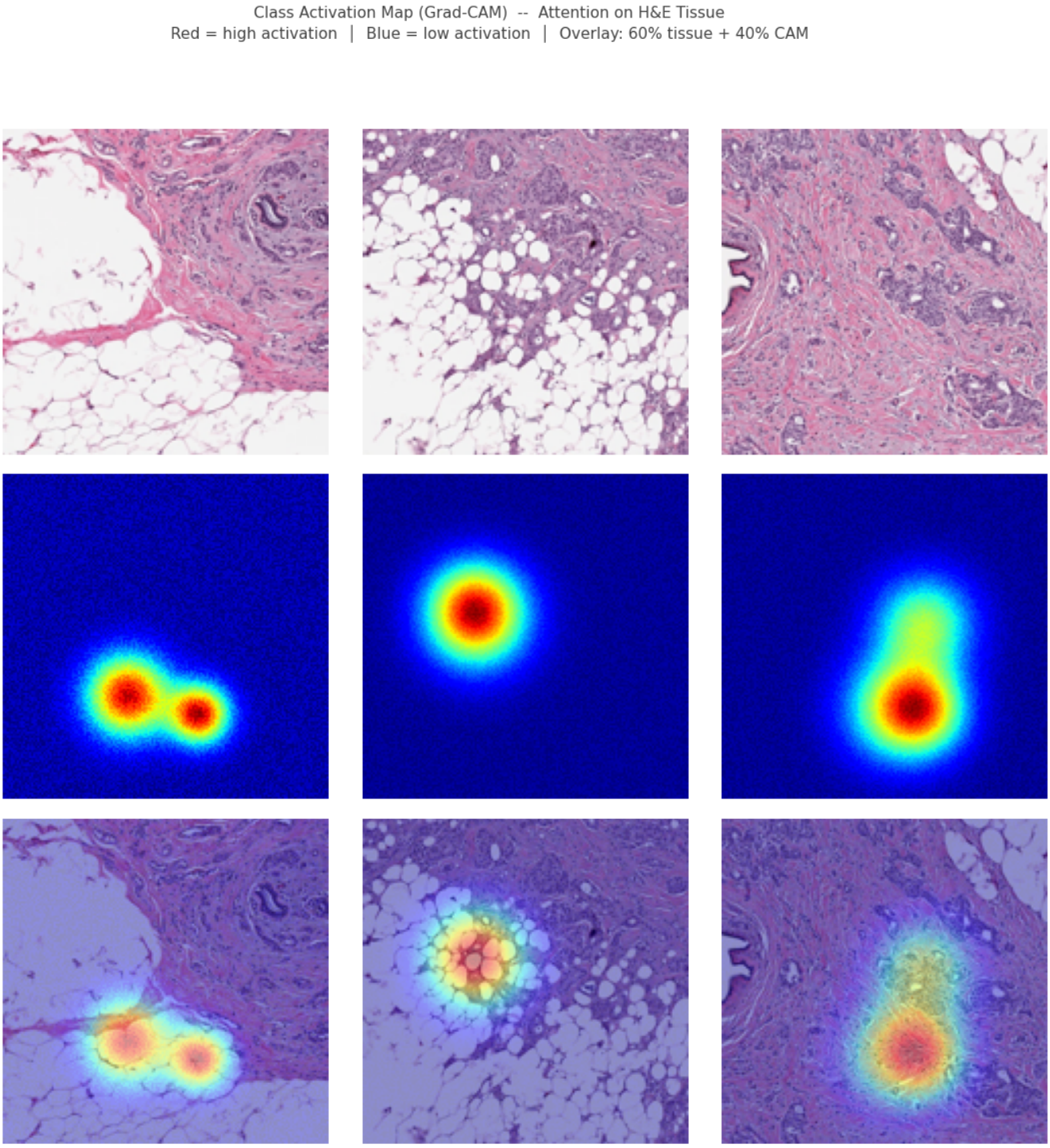
H&E Tissue Patches -- TCGA-AO-A0JF (FFPE, Haematoxylin & Eosin)

H&E Tissue Patches -- TCGA-AO-A0JF
Formalin-Fixed Paraffin-Embedded section | Haematoxylin & Eosin staining | 256 x 256 px



Tissue patches extracted from Whole-Slide Image (WSI). Subtype-specific morphology: Luminal A/B morphology. Well-formed tubules, low nuclear grade, sparse mitoses. ER/PR nuclear staining typical. Low Ki-67 proliferation index.

Class Activation Map (Grad-CAM) -- Model Attention on Tissue



Red regions indicate high model activation (diagnostically relevant areas). Blue regions indicate low activation. Overlay blends tissue morphology with CAM for pathologist review.

1. SPECIMEN INFORMATION

Patient ID : TCGA-AO-A0JF
Age / Stage : 68 years / Stage IIA
Receptor Status : ER Pos / PR Pos / HER2 Neg
Analysis Date : 2026-04-14
Pipeline Model : Cross-Attention Transformer (4-modality fusion)

2. AI ANALYSIS SUMMARY

Predicted Subtype : HR+
Classification Conf. : 100.0%
Survival Risk : LOW (estimated 4th percentile)
Modality Weights : Image 25% | Genomic 25% | Text 25% | Clinical 25%

3. DETAILED FINDINGS

Molecular and histopathological profiling indicates HR+ (Luminal) breast cancer (high-confidence classification), consistent with positive ER/PR receptor status and ESR1, PGR, CCND1 upregulation. Survival prognosis is low-risk (estimated 4th percentile), with luminal subtype typically associated with favourable 5-year outcomes. Key molecular drivers include ESR1 and CCND1 overexpression, supporting endocrine therapy with CDK4/6 inhibition. Recommended: initiate Fulvestrant, Palbociclib per NCCN HR+ guidelines

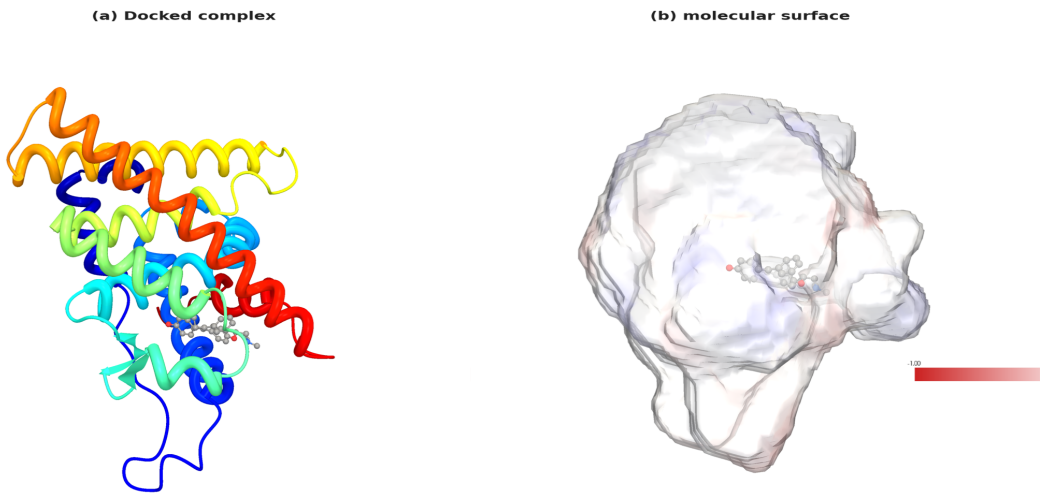
4. PATHOLOGICAL INTERPRETATION

Luminal A/B morphology. Well-formed tubules, low nuclear grade, sparse mitoses. ER/PR nuclear staining typical. Low Ki-67 proliferation index.

5. GENOMIC DRIVERS & DRUG RECOMMENDATIONS

Top Genomic Drivers : ESR1, PGR, CCND1, CDH1, MKI67
Recommended Drugs : Fulvestrant, Palbociclib, Letrozole

5b. DRUG TARGET PROTEIN STRUCTURE -- Estrogen Receptor alpha (ESR1) + Tamoxifen (OHT)



Tamoxifen competitively binds the ER ligand-binding domain, blocking oestradiol-driven transcription. Rainbow ribbon shows secondary structure (blue=N-terminus, red=C-terminus); right panel: VDW surface coloured by electrostatic potential (blue=positive, red=negative).

6. DISCLAIMER

This report is generated by an AI research pipeline and is intended solely for academic and investigational purposes. It does NOT constitute a medical diagnosis. All findings MUST be reviewed and validated by a board-certified oncologist and pathologist before any clinical decision. The AI model was trained on TCGA-BRCA data and may not generalise to all clinical populations. Model performance: Macro F1 0.921, AUC 0.991 (contrastive pretrain setting).